Diffraction

antocciphysics | TEACHER KEY | CPA

Learning target: Explain spreading around edges and through openings.

Suggested first pass: 1, 2, 3, 7. Add independent, mixed and stretch work from the bank as time and student evidence permit. Estimated times are planning estimates, not completion requirements.

01. Bring back an earlier idea (retrieve) *Review: Unit 6, Day 3; 3 course days later.*
24 m/s from $v=f\lambda$.

02. Explain a worked example (worked)
Strong curved spreading from narrow opening; mostly forward wavefronts with edge spreading from wide opening.

03. Look and explain (model)
Narrow opening produces strong spreading; wide opening mostly forward with edge effects.

04. Build the solution (practice)
A, width comparable toλ.

05. Use the idea (reason)
Edges/openings cause spreading within same medium; refraction involves speed change and boundary/gradient.

06. Choose a method (mixed) *Review: Unit 5, Day 6; 12 course days later.*
Same average per particle; total translational energy is 4 times as large.

07. Exit check (exit)
More, relative to the opening.

08. STRETCH: Apply it another way (stretch)
Usually less pronounced spreading when wavelength is smaller relative to opening.

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01. Bring back an earlier idea (retrieve) *Review: Unit 6, Day 3; 3 course days later.*
v=30 m/s; λ=30/50≈0.6 m.

02. Explain a worked example (worked)
Strong curved spreading from narrow opening; mostly forward wavefronts with edge spreading from wide opening.

03. Represent and interpret (model)
Narrow opening produces strong spreading; wide opening mostly forward with edge effects.

04. Solve and justify (practice)
Soundλ1 m comparable to doorway; lightλtiny relative opening, so far less noticeable diffraction at that scale.

05. Analyze the reasoning (reason)
Edges/openings cause spreading within same medium; refraction involves speed change and boundary/gradient.

06. Choose a method (mixed) *Review: Unit 5, Day 6; 12 course days later.*
Rms speed becomes 4 times as large because mean kinetic energy is proportional to the square of rms speed.

07. Exit check (exit)
λhalves; less pronounced spreading when opening is large compared with newλ.

08. STRETCH: Apply it another way (stretch)
Wavelengths 2 and 0.2 m; 170 Hz has greater wavelength/opening ratio and stronger spreading.